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(1) Research Program/Project/Study Title: Osmotic Dehydration of Kamias (<i>Averrhoa bilimbi</i>): Optimized Formulation, Physicochemical and Sensory Properties	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Mambusao
Province:	Capiz
Region:	VI

(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity Food/ Kamias	(8) Discipline/Field of Specialization Food Science and Technology
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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Kamias (Averrhoa bilimbi Linn.) is one of the important medicinal plants of many tropical and subtropical countries of the world which has been widely used in the traditional system of medicines for the treatment of a variety of ailments, particularly as an antidiabetic, antihypertensive, and antimicrobial agent. However, this fruit is oftentimes unappreciated and is allowed to fall from the tree and rot on the ground. Generally, this study aimed to develop and characterize the physico-chemical and sensory properties of dried kamias. The specific objectives are the following: to evaluate the sensory properties of osmotically dehydrated kamias; to determine the physico-chemical characteristics of osmotically dehydrated kamias; to evaluate the colorimetric properties of osmotically dehydrated kamias; to optimize the level of baking soda and syrup concentration of osmotically dehydrated kamias and lastly to determine the cost of production of osmotically dehydrated kamias. Results of the study revealed that dried kamias soaked in 5g baking soda solution and 40°B syrup concentration had the highest color acceptability while dried kamias soaked in 15g baking soda solution and 40°B syrup concentration had the highest aroma, taste, texture, and general acceptability; Likewise, dried kamias soaked in 15g baking soda solution and 40°B and 50°B syrup concentration also had the highest general acceptability rating. Further kamias soaked in 15g baking soda solution and 60°B syrup concentration had the highest aftertaste acceptability. In terms of physicochemical properties, pH was not significantly different among treatments. Total Soluble Solids (TSS) revealed a significant effect on the variable considered this means that the increase in the levels of baking soda and syrup concentration resulted in a significant effect on the TSS of dried kamias both linearly and quadratically while moisture content was highest when the levels of baking soda and syrup concentration were at the middle. For color analysis, dried kamias from 5g baking soda and 40°B syrup concentration had the highest browning and yellowness indices with a value of 3.43 and 4.50, respectively. The optimum formulation for dried kamias was 15 g baking soda and 40°B syrup concentration.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Baking soda, Kamias, Osmotic dehydration, syrup concentration, optimization

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Averrhoa bilimbi Linn. is one of the important medicinal plants of many tropical and subtropical countries of the world which has been widely used in the traditional system of medicines for the treatment of a variety of ailments, particularly as an antidiabetic, antihypertensive, and antimicrobial agent. *Averrhoa bilimbi* has many names in the Philippines, such names include: iba, ibag, ibe, kalmias, kalanuas, kalinguia, kolonanas, kolonauasm killing-iba, pias, and puis (Castro, 2021). It belongs to the family Oxalidaceae and typically found in tropical countries. It can grow up to 15 meters high and its fruits are relatively cylindrical and developed in bunches. When it reaches the maturity stage, the fruits' color changes from green to light yellow. This fruit has a juice yield of 76.14% and a very sour taste which can be utilized in pickles, wine, and vinegar but can also be eaten raw or dipped in rock salt (Caoli et al, 2017). When the fruit is already matured, it can be processed into jams

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and wine. The sour fruit is a natural source of vitamins B and C, iron, phosphorus and antioxidants. Its fruits can control gastrointestinal bleeding in the stomach, fight cholesterol, fever, and inflammation, and prevent cough, beriberi and scurvy (Valencia and Bismark, 2015). This fruit is oftentimes taken for granted and unappreciated hence they are allowed to fall from the tree and rot on the ground, however, it has the potential to be processed and used as a food product (Nunes et al, 2021).

Furthermore, osmotic dehydration is the phenomenon of removal of water from a lower concentration of solute to a higher concentration through semi permeable membrane that results in the equilibrium condition in both sides of membrane (Tiwari 2005; Yadav and Singh, 2014, Haneefa, 2023). In osmotic dehydration the solutes used are generally sugar syrup with fruit slices or cubes and salt (sodium chloride) or brine with fruits and vegetables. In osmotic dehydration using sugar as the osmotic agent, the sugar is added periodically to maintain the concentration level usually 70 °B (Yadav et al., 2014). This is a multicomponent diffusion process. In this process, water flows from fruits or vegetables to solution and along with water some components of fruits and vegetables such as minerals, vitamins, fruit acids, etc. also move towards the solution. The sugar and salt migrate towards the fruits and vegetables (Yadav and Singh, 2014). This osmotic dehydration technique reduces production cost, production time and also improves the quality of the product and prolong the shelf life (Sriraaman et al, 2021). Hence, this study was conducted.

Objectives of the Study:

Generally, this study aimed to develop and characterize the physico-chemical and sensory properties of dried kamias.

The specific objectives are the following:

1. To evaluate the sensory properties of osmotically dehydrated kamias.
2. To determine the physico-chemical characteristics of osmotically dehydrated kamias.
3. To evaluate the colorimetric properties of osmotically dehydrated kamias.
4. To optimize the level of baking soda and syrup concentration of osmotically dehydrated kamias.

(11) Materials and Methods or Methodology

Materials

The materials used in the conduct of the study were the following: kamias, refined sugar, distilled water, baking soda, knife, measuring cylinder, plate, mixing bowl, gas stove, spoon, strainer, plastic wrap, stirring spoon, weighing scale, digital weighing scale, food dehydrator, and foil laminated pouches.

Methods

Experimental Treatment

The study followed a 3x3 factorial experiment. Factor A was the level of Baking Soda while factor B was the level of syrup concentration. Factor A was further composed of A1- 5 g/ L solution, A2- 10 g/L and A3-15g/L. Similarly, Factor B was composed of B1- 40°B, B2- 50 °B and B3-60 °B .

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There were a total of nine (9) experimental treatments, namely:

- T1- 5g baking soda at 40°B syrup concentration
- T2- 10g baking soda at 40°B syrup concentration
- T3- 15g baking soda at 40 °B syrup concentration
- T4- 5g baking soda at 50 °B syrup concentration
- T5- 10g baking soda at 50 °B syrup concentration
- T6- 15g baking soda at 50 °B syrup concentration
- T7- 5g baking soda at 60 °B syrup concentration
- T8- 10g baking soda at 60 °B syrup concentration
- T9- 15g baking soda at 60 °B syrup concentration

Preparation of Dried Kamias

1. Select mature fruit and of good quality kamias.
2. Cut the ends of the fruit to facilitate absorption of the soaking solution and to remove tiny hair-like structure at the ends.
3. Soak in Baking soda solution for 24 hours.
4. After 24 hours, drain and wash the kamias fruit.
5. Soak in syrup of different concentrations corresponding to assigned treatment.
6. Check the daily syrup concentration using a handheld refractometer for seven (7) days.
7. If the syrup decreased its concentration due to osmotic dehydration, add sugar until the desired concentration is attained.
8. Drain the soaked kamias in syrup every other day. Heat the syrup in low fire for 5 minutes. Allow to cool and soak again the kamias fruit. Repeat the process up to 7 days.
9. After seven days, drain the kamias until no syrup drips from the fruit.
10. Arrange in drying trays and dehydrate at 43°C for 10 hours.
11. Cool the dried kamias and pack in foil laminate pouch.
12. Label and store

Sensory Evaluation

Dried kamias as affected by varying levels of baking soda and syrup concentration were subjected to sensory evaluation to determine which treatment is acceptable in terms of color, aroma, taste, texture, aftertaste, and general acceptability using the 9-point Hedonic Scale. The quality scoring method was used to describe the attributes of the finished product (Hilapad et al, 2020).

Physico-Chemical Analysis of Dried Kamias

The pH and Total Soluble Solid (TSS) of kamias candy as affected by varying levels of baking soda and sugar were determined with the use of a pH meter and a handheld refractometer, respectively. Moisture content (%MC) was determined using a moisture meter.

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Color Analysis of Dried Kamias

The Hunter L*, a*, and b* values of meat loaf were measured using a colorimeter. The dimension L* means lightness, a* value indicates redness when positive and greenness when negative and the b* value represents yellowness when positive and blueness when negative. Browning index was calculated following the formula of Mohapatra et al (2010):

$$BI = 100 \times \frac{X-0.31}{0.17}$$

Where:

$$X = \frac{a^* + 1.75L^*}{(5.645L^* + a^* - 3.012b^*)}$$

Yellowness index was calculated following the formula:

$$YI = 142.86b^*/L^*$$

(12) Results and Discussions

Sensory Evaluation

Product Acceptability

Dried kamias made from different treatments were subjected to sensory evaluation to determine their acceptability in all the attributes considered such as color, aroma, taste, texture, aftertaste, and general acceptability. The mean acceptability ratings for sensory attributes of dried kamias as affected by varying levels of baking soda and syrup concentration are presented in Table 1. The Table shows that color acceptability ranged from 7.38 to 8.0 and falls within the like moderately to like very much on the 9-point hedonic scale. Dried kamias which was soaked in 5 g/ L baking soda solution and 40°B syrup concentration had the highest color acceptability of 8.0 and fell in the like very much category. In terms of aroma, acceptability scores ranged from 7.29 to 7.71 with dried kamias soaked in 15g baking soda and 40°B syrup concentration obtaining the highest aroma acceptability rating of 7.71. For taste, texture, aftertaste and general acceptability, the acceptability ratings range were 7.42 to 7.88, 7.25-7.67, 7.42 to 7. 88 and 7.58 to 7.92, respectively. All the acceptability ratings were within the like moderately to like very much category in the 9-point Hedonic scale. However, statistical analysis revealed no significant differences among variables under study. This implies that the level of baking soda and syrup concentration did not create a significant difference on the sensory acceptability of dried kamias. Studies conducted by Prasannath and Mehendran (2009) in osmotically dehydrated jackfruit and Nadia et al.,(2013) in pear revealed the positive effect of osmotic dehydration on sensory qualities.

Furthermore, Figure 1 presents the response surface plots for all the sensory attributes of dried kamias as affected by varying levels of baking soda and syrup concentration. The figure shows that color acceptability was highest when the level of syrup concentration was lowest across the lowest level of baking soda while aroma acceptability increased when the level of syrup concentration was lowest and the level of baking soda was highest. Likewise, taste

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acceptability increased when the level of syrup concentration was lowest across the highest level of baking soda. For texture acceptability, it increased when both levels of syrup concentration and baking soda were highest while aftertaste acceptability was highest when the levels of syrup concentration was both lowest and highest (40 and 60) across highest level of baking soda. Finally, for general acceptability, it was highest when the level of baking soda was highest across low and high levels of syrup concentration.

Product Description

Quality description of dried kamias as presented in Table 2 shows that the color of dried kamias varies from light yellow to yellowish brown, greenish brown, and brown. In terms of aroma, it ranged from slightly perceptible to moderately perceptible aroma while taste vary from slightly sweet to well-blended sweet and sour. The majority of the treatments were described as well-blended sweet and sour except for treatments 5g and 15 g baking soda and 50°B syrup concentration which were described as slightly sweet. Texture is a trait or a physical condition and morphology of agricultural products which includes the level of hardness, tenderness, flexibility, elasticity, roughness, and smoothness of materials (Buntaran et al, 2010). The texture of dried kamias was described as moderately hard and gummy except for treatments with 60°B syrup concentration which was described as just right. This means that as the syrup concentration increased, the texture softened thereby producing a dried kamias with just right description. Lastly, aftertaste description ranged from just right to moderately acidic taste. The reduction in acidic taste was due to the addition in baking soda during soaking.

Physico-chemical Properties of Dried Kamias

The physiochemical properties of foods are indicators of food quality, sensory, and safety. Understanding the physiochemical characterizations of foods is not only essential for food preservation and food quality assessment but also important for consumers' health.

The Physicochemical Characteristics of dried kamias as affected by varying levels of baking soda and syrup concentration are presented in Table 3. The data show that the pH of dried kamias ranged from 3.88 to 4.33. As shown in the Table, pH increased when the amount of baking soda was increased. This is because the addition of baking soda neutralizes the acidity of kamias hence increasing the pH level. The decrease in acidity during soaking in syrup was due to the bio-conversion of acids into sugars (Asparna et al, 2022; Lal *et al.*, 2015). Statistical analysis of the pH of dried kamias revealed no significant differences among treatment means. This implies that the levels of baking soda and syrup concentration did not significantly altered the pH of the finished product.

Furthermore, the Total Soluble Solids (TSS) revealed a significant effect on the variable considered this means that the increase in the levels of baking soda and syrup concentration resulted in a significant effect on the TSS of dried kamias both linearly and quadratically. It further reveals that the interaction effect between the levels of baking soda and syrup concentration was significant.

Similarly, the moisture content of dried kamias ranged from 5.63% to 9.42% with dried kamias soaked in 5g baking soda solution and 40°B syrup concentration having the lowest moisture

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content. Statistical analysis showed that the levels of baking soda significantly affected the moisture content of dried kamias both linearly and quadratically.

Color Analysis

Table 4 presents the color analysis of kamias as affected by varying levels of baking soda and syrup concentration. Statistical analysis revealed no significant differences in the L,a and b values of dried kamias from different treatments. The Table shows further that dried kamias with 5g baking soda and 40⁰B syrup concentration had the highest Browning index (BI) and Yellowness index (YI). Correlating this result with the acceptability rating for color, this implies that the increase in the BI and YI values enhanced the color acceptability.

Optimum Formulation

Parameters considered for optimum formulation were the acceptability rating of all the parameters evaluated. To further limit the optimum region, a cut-off acceptability score of 7.5 was set. The shaded region corresponds to the optimum formulation considering its acceptability among panelists. In this study, the superimposed contour plots revealed that the optimum region for dried kamias was at 15g baking soda and 40⁰B syrup concentration.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions:

Based on the findings of the study, the researcher has drawn the following conclusions:

1. Dried kamias with 5 g baking soda solution and 40⁰B syrup concentration had the highest color acceptability while dried kamias soaked in 15g baking soda and 40⁰B syrup concentration obtained the highest aroma, taste, texture, and general acceptability
2. The pH of dried kamias was not significantly affected; Total Soluble Solids (TSS) were significantly affected by the variable considered while moisture content was highest when the levels of baking soda and syrup concentration were at the middle.
3. Colorimetric properties showed no significant difference. Browning and yellowness indices were highest at 5 g baking soda and 40⁰B syrup concentration.
4. The optimum formulation was 15 g baking soda and 40⁰B syrup concentration

Recommendations

1. Use 15g baking soda and 40⁰B syrup concentration in the osmotic dehydration of kamias.
2. Conduct shelf-life testing of the optimized product.
3. Conduct a proximate analysis of the optimized product to comply with labeling standards and to determine its nutritional content.

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4. Determine the number of cycles for reusing the syrup for osmotic dehydration

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

1 Publication article drafted
1 UM application for filing
1 Product
1 trained adopter

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

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APPENDICES

Appendix 1: Tables

Table 1.0 Mean Acceptability ratings for sensory attributes of dried kamias as affected by varying levels of baking soda and syrup concentration

Treatment	Level of BS (g)	Syrup concentration (⁰ B)	Color ns	Aroma ns	Taste ns	Texture ns	Aftertaste ns	Gen. Acceptability ns
1	5	40	8.00	7.54	7.58	7.50	7.58	7.71
2	10	40	7.54	7.54	7.54	7.42	7.71	7.92
3	15	40	7.58	7.71	7.88	7.67	7.83	7.92
4	5	50	7.58	7.33	7.50	7.42	7.54	7.75
5	10	50	7.38	7.29	7.50	7.25	7.42	7.58
6	15	50	7.38	7.58	7.75	7.54	7.63	7.92
7	5	60	7.50	7.38	7.42	7.38	7.42	7.75
8	10	60	7.54	7.54	7.79	7.67	7.63	7.92
9	15	60	7.50	7.46	7.63	7.67	7.88	7.88

BS- Baking soda

ns- not significant at $p \leq 0.05$

Table 2. Quality description of of dried kamias as affected by varying levels of baking soda and syrup concentration

Treatment	Level of BS (g)	Syrup conc. (⁰ B)	Color	Aroma	Taste	Texture	Aftertaste
1	5	40	Brown	Slightly perceptible kamias aroma	Well blended sweet and sour	Moderately hard and gummy	Just right
2	10	40	Brown	Slightly perceptible kamias aroma	Well blended sweet and sour	Moderately soft and gummy	Slightly acidic aftertaste
3	15	40	Greenish brown	moderately perceptible kamias aroma	Well blended sweet and sour	Moderately soft and gummy	Slightly acidic aftertaste
4	5	50	Brown	moderately perceptible kamias aroma	Slightly sweet	Moderately soft and gummy	Just right
5	10	50	Greenish brown	Slightly perceptible kamias aroma	Well blended sweet and sour	Moderately soft and gummy	Absence of acidic aftertaste
6	15	50	Brown	Slightly perceptible kamias aroma	Slightly sweet	Moderately soft and gummy	Slightly acidic aftertaste

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7	5	60	Light yellow	moderately perceptible kamias aroma	Slightly sweet	Just right	Just right
8	10	60	Yellowish brown	moderately perceptible kamias aroma	Slightly sweet	Moderately soft and gummy	Moderately acidic aftertaste
9	15	60	Greenish brown	Slightly perceptible kamias aroma	Well blended sweet and sour	Just right	Just right

Table 3. Physicochemical Characteristics of dried kamias as affected by varying levels of baking soda and syrup concentration

Treatment	Level of BS (g)	Syrup conc. (%w/v)	pH	TSS(°B)	% MC
1	5	40	3.88	32.73	6.60
2	10	40	3.72	32.97	8.31
3	15	40	4.04	36.97	7.83
4	5	50	4.16	33.57	8.06
5	10	50	4.22	30.57	9.42
6	15	50	3.94	33.80	6.30
7	5	60	3.95	35.03	5.63
8	10	60	4.33	34.90	7.16
9	15	60	4.10	32.73	6.36

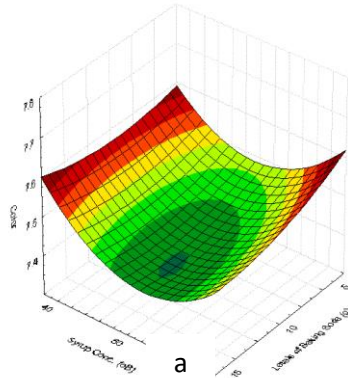
Table 4. Color Analysis of dried as affected by varying levels of baking soda and syrup concentration

Treat ment	Level of BS (g)	Syrup conc. (°B)	L	a	b	BI	YI
1	5	40	89.55	0.37	2.82	3.43	4.50
2	10	40	89.73	0.37	2.68	3.26	4.27
3	15	40	89.67	0.34	2.63	3.18	4.20
4	5	50	89.71	0.40	2.64	3.23	4.20
5	10	50	89.72	0.37	2.67	3.25	4.26
6	15	50	89.72	0.33	2.64	3.17	4.20
7	5	60	89.71	0.37	2.66	3.24	4.24
8	10	60	89.65	0.35	2.65	3.21	4.22
9	15	60	89.47	0.31	2.79	3.34	4.45

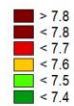
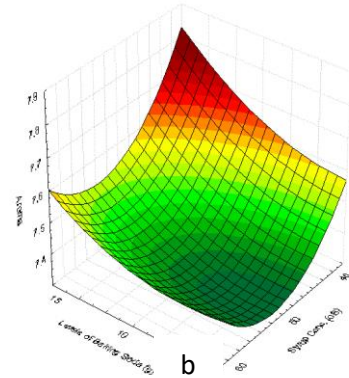
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Appendix 1: Figures

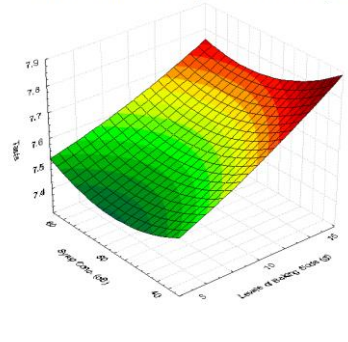
$$\text{Color} = 9.5 - 2.1511E-15x - 0.0806y + 0.0008x^2 - 0.0004x^2y + 0.0008y^2$$



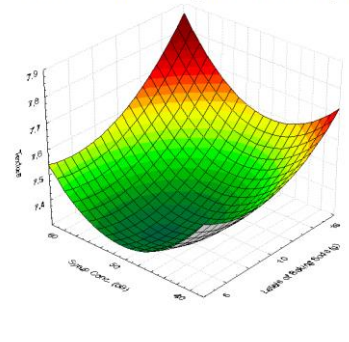
$$\text{Aroma} = 10.6389 + 0.0042x - 0.1278y + 0.0017x^2 - 0.0004x^2y + 0.0012y^2$$



$$\text{Taste} = 8.6991 + 0.0347x - 0.0542y + 0.0006x^2 - 0.0004x^2y + 0.0006y^2$$



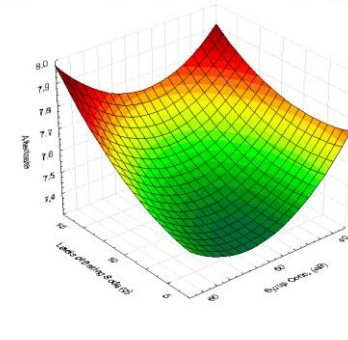
$$\text{Texture} = 11.3403 - 0.0785x - 0.15y + 0.0033x^2 + 0.0006x^2y + 0.0015y^2$$



c

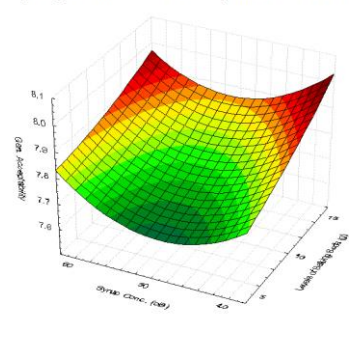
d

$$\text{Aftertaste} = 11.8125 - 0.0757x - 0.1597y + 0.0025x^2 + 0.001x^2y + 0.0015y^2$$



e

$$\text{Gen. Acceptability} = 9.8519 + 0.0264x - 0.0931y + 0.0006x^2 - 0.0004x^2y + 0.001y^2$$



f

Figure 1. Response surface plot for sensory acceptability of dried kamias as affected by varying levels of baking soda and syrup concentration

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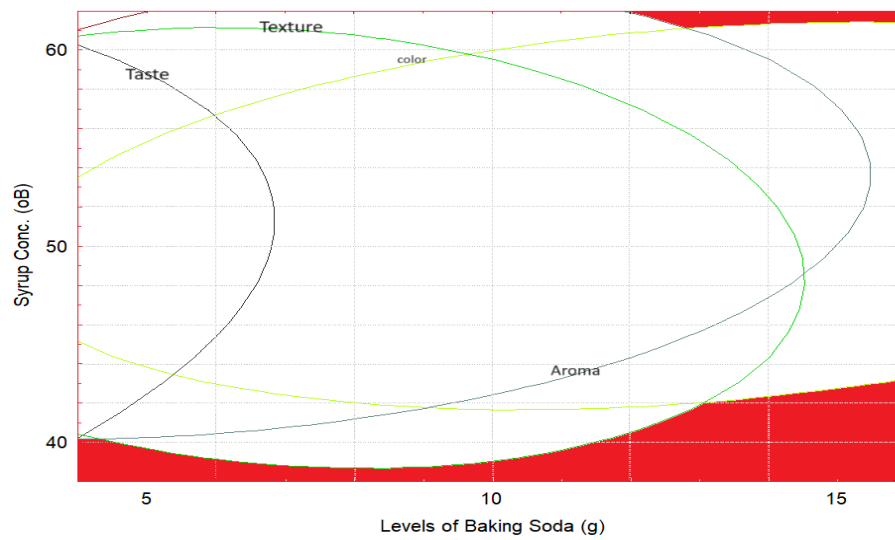


Figure 2. Optimum region of as affected by varying levels of baking soda and syrup concentration